

UNIT II

2 Marks

Define Software. Software is a generic term for organized collection of computer data and instructions. It is responsible for controlling, integrating, and managing the hardware components of a computer and to accomplish specific tasks. In other words, software tells the computer what to do and how to do it.

Define System Software. Give example.

- ❖ System software consists of several programs, which are directly responsible for controlling, integrating, and managing the individual hardware components of a computer system
- ❖ System software acts as an interface between the hardware of the computer and the software applications.
- ❖ Some examples of system software are operating systems, device drivers, language translators, and system utilities.

Define Application Software. Give example.

- ❖ The software that a user uses for accomplishing a specific task is the application software.
- ❖ Application software may be a single program or a set of programs
- ❖ Application software is written for different kinds of applications—graphics, word processors, media players, database applications, telecommunication, accounting purposes etc.
- ❖ Some of the most commonly used application software are: Word Processors, Spreadsheets

Define Utility Software? List any two System Utility functions.

- ❖ System utility programs perform day-to-day tasks related to the maintenance of the computer system.
- ❖ They are used to support, enhance, expand, and secure existing programs and data in the computer system.
- ❖ Most common functions of system utilities include: File Management, Backup, Data Recovery, Virus Protection, Disk Management, Firewalls, Disk Cleanup

Differentiate compilers and interpreters

Compiler is a language processor that converts a program written in high-level language into machine language, entire program at once, is called a compiler. Thus, the input of a compiler is a high-level language code (called source code), while its output is a machine language code (called object code). The common programming languages that use compiler are C, C++, C#, etc.

Interpreters is language translator that converts a high-level language program into a machine language program, one line at a time, is referred to as an interpreter. The programming languages that use interpreters are Perl, Ruby, Python, METLAB, etc.

What is Machine Language The first language was binary, also known as machine language, which was used in the earliest computers and machines. We know that computers are digital devices, which have only two states, ON and OFF (1 and 0). Therefore, every instruction and data should be written using 0s and 1s. Machine language is also known as the computer's "native" language as this system of codes is directly understood by the computer.

What is Assembly Language An assembly language provides a mnemonic instruction, usually three letters long, corresponding to each machine instruction. The letters are usually abbreviated indicating what the instruction does. For example, ADD is used to perform an addition operation, SUB for subtraction, and so on.

What is Loader

- ❖ Loaders are a part of the operating system that brings an executable file residing on disk into memory and starts its execution. It is responsible for loading, linking and relocation.
- ❖ It is responsible for loading, linking and relocation.
- ❖ In computing, a loader is a program that performs the functions of a linker and then immediately schedules the executable code for execution, without necessarily creating an executable file as an output.

List basic tasks of a loader.

A loader performs four basic tasks as follows.

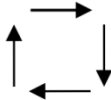
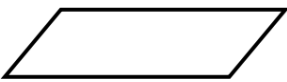
- ❖ **Allocation:** It allocates memory space for the programs.
- ❖ **Linking:** It combines two or more separate object programs and supplies the information needed to allow references between them.
- ❖ **Relocation:** It prepares a program to execute properly from its storage area.
- ❖ **Loading:** It places data and machine instructions into the memory.

Define Algorithm Algorithms are step-by-step procedures designed to solve specific problems and perform tasks efficiently in the problem solution of computer science and mathematics. These powerful sets of instructions form the backbone of modern technology and govern everything from web searches to artificial intelligence.

Define flowcharts A flowchart is a pictorial representation of an algorithm in which the steps are drawn using different geometric shapes. Each geometric shape represents an operation. The logical flow is indicated by interconnecting arrows, which represents the sequence in which the operations are implemented. Flowcharts outline the general procedure and provide an alternative, visual way of representing the information flow in

a program. The primary purpose of the flowchart is to help the programmer in understanding the logic of the program.

List any 4 symbols used in flow charts along with their purpose

Symbol	Symbol Name	Description
	Flow Lines	Flow lines are used to connect symbols. These lines indicate the sequence of steps and the direction of flow of control.
	Terminal	This symbol is used to represent the beginning (start), the termination (end) or halt (pause) in the program logic.
	Input/Output	This symbol represents information entering or leaving the system such as customer order (input) and servicing (output).
	Processing	This symbol is used for representing arithmetic and data movement instructions. It can represent a single step ("add two cups of flour"), or an entire sub-process ("make bread") within a larger process.

What is pseudocode

- ❖ Pseudocode is an outline of a program, written in a form that can be easily converted into actual programming statements.
- ❖ Pseudocode uses plain English statements rather than symbols to represent the processes in a computer program. It is also known as PDL (program design language), as it emphasizes more on the design aspect of a computer program or structured English, because usually pseudocode instructions are written in normal English, but in a structured way.

List any two limitations of pseudocode.

- ❖ It does not provide a visual representation of the program's logic.
- ❖ There are no accepted standards for writing pseudocodes. Programmers use their own style of writing pseudocode. It is quite difficult for the beginners to write pseudocode as compared to drawing a flowchart.

What is Word Processor? Give example for Word Processors

- ❖ A word processor is a software application used for creating, editing, formatting, and printing text-based documents.
- ❖ It involves not only typing, but also checking the spelling and grammar of the text and arranging it correctly on the page.
- ❖ It is possible to include pictures, graphs, charts, and many other things within the text of the document. It also allows for changes in margins, fonts, and colour.
- ❖ Ex: Microsoft Word, Google Docs.

What is Spreadsheet Software Give example A spreadsheet application is a rectangular grid, which allows text, numbers, and complex functions to be entered into a matrix of thousands of individual cells. The spreadsheet provides sheets containing cells, each of which may contain text and/or numbers. Ex: Microsoft Excel, Google Sheets

What are image editors? Give example

- ❖ Image editor programs are designed specifically for capturing, creating, editing, and manipulating images. A word processor is a software application used for creating, editing, formatting, and printing text-based documents.
- ❖ With image editing software, one can darken or lighten an image, rotate it, adjust its contrast, crop out extraneous detail, and much more.
- ❖ Ex: Adobe Photoshop, Coral Draw

What is Database Management System? Give example

- ❖ Database management software is a collection of computer programs that allow storage, modification, and extraction of information from a database in an efficient manner.
- ❖ It supports the structuring of the database in a standard format and provides tools for data input, verification, storage, retrieval, query, and manipulation.
- ❖ Ex: MS-Access, Oracle

What is Presentation Software? Give example for Presentation Software

- ❖ A presentation is a means of expressing ideas in the form of visual images, audio, and animation. To prepare presentations, presentation software is used.
- ❖ Presentation software allows the users to create presentation by producing slides or handouts for presentation of projects. Essentially, such computer programs allow users to create a variety of visually appealing electronic slides for presentations. Microsoft PowerPoint is one of the most famous presentation applications.
- ❖ Ex: Microsoft PowerPoint, Google Slides

What is Desktop Publishing? Give example for Desktop Publishing Software.

- ❖ The term desktop publishing is usually used to describe the creation of printed documents using a desktop computer. It is a technique of using a personal computer to design images and pages, assemble type and graphics, and then using

a laser printer or image setter to output the assembled pages onto a paper, film, or printing plate.

- ❖ Desktop publishing software is used for creating magazines, books, newsletters, and so on. Such software assists in creating sophisticated documents including complicated page designs, detailed illustrations, and camera-ready typefaces. Quark Express and Adobe PageMaker are desktop publishing software.
- ❖ Ex: Adobe PageMaker, Microsoft Publishers

What is Linker. Differentiate Linker and Loader.

A **Linker** is a system program that links together several objects, modules, and libraries to form a single, coherent program (executable program).

Linker	Loader
The main function of Linker is to generate executable files.	Whereas main objective of Loader is to load executable files to main memory.
The linker takes input of object code generated by compiler/assembler .	And the loader takes input of executable files generated by linker.

Differentiate Absolute Loader and Relocating Loader

Absolute Loader	Relocating Loader
It loads the file into memory at the location specified by the beginning portion (header) of the file and the passes control to the program.	This loader loads the program in memory.

5M, 6M

Define Operating System. List and explain basic functions of an Operating System (6)

- ❖ It is the first layer of software loaded into the computer memory when it starts up.
- ❖ As the first software layer, all other software that gets loaded after it depends on it for various common services. These common core services include disk access, memory management, task scheduling, and user interfacing
- ❖ Operating system organizes and controls the hardware.
- ❖ In addition, the operating system ensures that different programs executing at the same time do not interfere with each other.
- ❖ Examples of Operating Systems are Windows, Linux, macOS, Android, iOS, etc.

The basic functions of an operating system are:

- ❖ **Process Management:** It handles the creation, deletion, suspension, resumption, and synchronization of processes.
- ❖ **Memory Management:** It handles allocation and de-allocation of memory space as required by various programs.
- ❖ **File Management:** It is responsible for the creation and deletion of files and directories. It also organizes, stores, retrieves, names, and protects all the files.
- ❖ **Device Management:** It manages all the devices of the computer system such as printers and modems. If any device fails, it detects the device failure and notifies the same to the user.
- ❖ **Security Management:** It protects system resources and information against destruction and unauthorized use.
- ❖ **User Interface:** It provides the interface between the user and the hardware.

Write a note on

a. Device Drivers

- ❖ Device drivers are system programs, which are responsible for proper functioning of devices.
- ❖ Every device, whether it is a printer, monitor, mouse, or keyboard, has a driver program associated with it for its proper functioning.
- ❖ A driver acts like a translator between the device and program (typically, an operating system) that uses the device.
- ❖ A device driver is not an independent program, it assists and is assisted by the operating system for the proper functioning of the device.

b. Language Translators

- ❖ Computers only understand machine language that is a language consisting of 0's and 1's.
- ❖ Hence with every programming language, a language translator is also developed, which accepts the programs written in the programming language and transforms them into a form suitable for execution.
- ❖ Language translators are divided in three major categories: **Compiler, Interpreter, and Assembler** - (write the definition all three Language translators)

What is System Utility? Explain any five functions of System Utility

- ❖ System utility programs perform day-to-day tasks related to the maintenance of the computer system.
- ❖ They are used to support, enhance, expand, and secure existing programs and data in the computer system.

Most common functions of system utilities include:

- ❖ **File Management:** These utilities make it easier to manage data files. Many programs are written to help users to find the files, create and organize directories, copy, move, and remove files. Example: File Manager in Windows, macOS Finder.
- ❖ **Backup:** Sometimes, it may happen that some data files are corrupted or accidentally deleted. In such a case, data backups become very useful. A backup system utility is essential for those organizations that want to keep their data intact.
- ❖ **Data Recovery:** It is the process of retrieving deleted or inaccessible data from failed electronic storage media such as computer hard disk drives, removable media, optical devices, and tape cartridges. Ex. Disk Drill
- ❖ **Virus Protection:** To protect system from viruses, one needs anti-virus software. Once installed on the system, anti-virus software scans the hard disk for any kind of virus and, if found, removes them.
- ❖ **Disk Management:** Disk management programs include various software like disk defragmenter, data compressor, and disk formatting tools.
 - **De-fragmentation** implies putting fragments of files in a sequential order onto the disk, which reduces the time to access the file.
 - **Data compression** programs squeeze out the slack space generated by the formatting schemes.
 - **Formatting tools** format the hard drive in tracks and sectors for orderly storing of data in the disk.
- ❖ **Firewalls:** Firewalls form a barrier between networked computers within an organization and those outside the organization. A firewall is designed to protect a computer from unauthorized access, especially via network.
- ❖ **Disk Cleanup:** It is utility tool to scan and delete unnecessary files from a computer's hard drive, such as temporary files, system files, and Recycle Bin contents, to free up storage space and improve system performance.

What is Application Software? Explain any two commonly used application software.

The software that a user uses for accomplishing a specific task is the application software. Application software may be a single program or a set of programs. Application software is written for different kinds of applications—graphics, word processors, media players, database applications, telecommunication, accounting purposes etc.

Some of the most commonly used application software are:

Word Processors:

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text of the document. It also allows for changes in margins, fonts, and colour. Ex: Microsoft Word, Google Docs.

- ❖ Ex: Microsoft Word and WordPerfect

Spreadsheets:

- ❖ A spreadsheet application is a rectangular grid, which allows text, numbers, and complex functions to be entered into a matrix of thousands of individual cells.
- ❖ The spreadsheet provides sheets containing cells, each of which may contain text and/or numbers. Cells may also contain equations that calculate the results from the data placed in other cells or a series of cells.
- ❖ Spreadsheet software is used predominantly for accounting and other numerically based tasks, because financial and mathematical data and analysis are often managed in a tabular style, with columns of financial figures being manipulated and then influencing other data computations.
- ❖ Ex: Microsoft Excel, Google Sheets

Image Editors:

- ❖ Image editor programs are designed specifically for capturing, creating, editing, and manipulating images. A word processor is a software application used for creating, editing, formatting, and printing text-based documents.
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Database Management Systems:

- ❖ Database management software is a collection of computer programs that allow storage, modification, and extraction of information from a database in an efficient manner.
- ❖ It supports the structuring of the database in a standard format and provides tools for data input, verification, storage, retrieval, query, and manipulation.
- ❖ It also controls the security and integrity of the database from unauthorized access.
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Presentation Applications:

- ❖ A presentation is a means of expressing ideas in the form of visual images, audio, and animation. To prepare presentations, presentation software is used.
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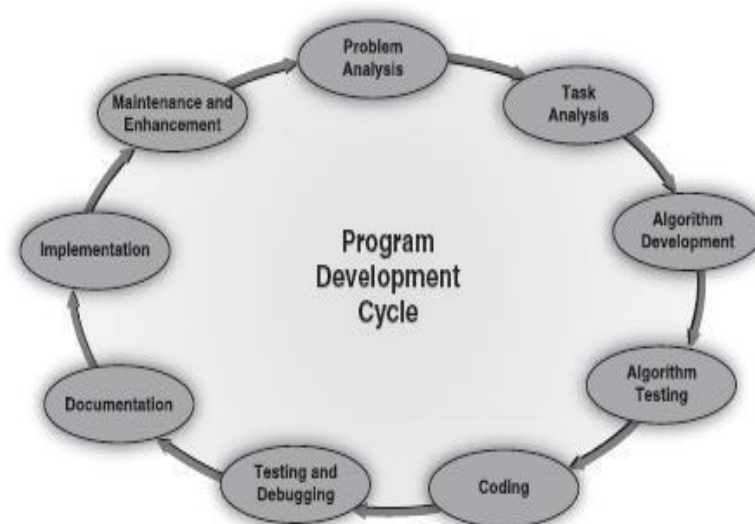
What are assemblers? List and explain different functions of Assemblers

- ❖ It is a system software which translates programs written in assembly language into machine language.
- ❖ Assembler converts the assembly codes into binary codes and then it assembles the machine understandable code into the main memory of the computer, making it ready for execution

Functions of an assembler are:

- ❖ It translates mnemonic operation codes to machine code and corresponding register addresses to system addresses.
- ❖ It checks the syntax of the assembly program and generates diagnostic messages on syntax errors.
- ❖ It assembles all the instructions in the main memory for execution.
- ❖ In case of large assembly programs, it also provides linking facility among the subroutines.
- ❖ It facilitates the generation of output on the required output medium.

Briefly explain different phases of the Program Development Cycle



- ❖ **Problem Analysis**
- ❖ **Task Analysis**
- ❖ **Algorithm Development**
- ❖ **Algorithm Testing**
- ❖ **Coding (HLL).**
- ❖ **Testing and Debugging**
- ❖ **Documentation**
- ❖ **Implementation**
- ❖ **Maintenance and Enhancement**

- ❖ **Problem Analysis:** Problem analysis is the process of understanding real-world problems. The goal of problem analysis is to gain a better understanding of the problem being solved, before development begins.
- ❖ **Task Analysis:** After analyzing the problem, the developer needs to develop various solutions to solve the given problem. From these solutions, the optimum solution is chosen.
- ❖ **Algorithm Development:** After selecting the appropriate solution, it is a process of designing step-by-step instructions that a computer follows to solve a problem. An algorithm is represented by flowcharts, decision tables and pseudocodes.
- ❖ **Algorithm Testing:** Before converting the algorithms into actual code, it should be checked for accuracy. The main purpose of checking the algorithm is to identify major logical errors at an early stage because logical errors are often difficult to detect and correct at later stages.
- ❖ **Coding:** The actual coding of the program takes place in the chosen programming language. A program can be written by using computer languages of different levels such as machine, assembly or high-level languages (HLL).
- ❖ **Testing and Debugging:** A program compiler tests the code for syntax errors. The results obtained are compared with results calculated manually from these test data. Depending upon the complexity of the program, several rounds of testing may be required.
- ❖ **Documentation:** Once the program is free from all the errors, it is the duty of the program developers to ensure that the program is supported by suitable documentation. These documents should be supplied to the program users. Documenting a program enables the user to operate the program correctly.
- ❖ **Implementation:** After documentation, the program is installed on the end user's machine and the user is also provided with all the essential documents in order to understand how the program works.

- ❖ **Maintenance and Enhancement:** After the program is implemented, it should be properly maintained by taking care of the changing requirements of its users and the system. This phase is also concerned with detecting and fixing the errors, which were missed in the testing phase. Since this step generates user feedback, the programming cycle continues as the program is modified or reconstructed to meet the changing needs.

List and explain the advantages & disadvantages machine level languages?

Advantages

- ❖ **Translation Free:** Machine language is the only language that computers can directly execute without the need for conversion.
- ❖ **High Speed:** Since no conversion is needed, the applications developed using machine language are extremely fast. It is usually used for complex application such as space control system, nuclear reactors and chemical processing.

Disadvantages

- ❖ **Machine Dependent:** Every computer type differs from the other, based on its architecture. Hence, an application developed for a particular type of computer may not run on another type of computer.
- ❖ **Complex Language:** Machine language is very difficult to read and write. Since all the data and instructions must be converted to binary code, it is almost impossible to remember the instructions.
- ❖ **Error Prone:** Since the programmer has to remember all the opcodes and memory locations, machine language is bound to be error prone. It requires a superhuman effort to keep track of the logic of the problem, and therefore results in frequent programming errors.
- ❖ **Tedious:** Machine Language programming becomes too complex to Modify and the programmer has to rewrite the entire logic again. Therefore, it is very tedious and time consuming, and since time is a precious commodity, programming using machine language tends to be costly.

List and explain the advantages & disadvantages Assembly level languages

Advantages

- ❖ **Easy to Understand and Use:** The programs written in assembly language are much more easier to understand and use as compared to its machine language counterpart. Being a more user- friendly language as compared to machine language, assembly programs are easier to Modify.

- ❖ **Less Error Prone:** There are fewer errors while writing an assembly language program. Even in case of errors, assembly programs provide better facility to locate and correct them as compared to machine language programs.
- ❖ **Efficiency:** Assembly programs can run much faster and use less memory and other resources than a similar program written in a high-level language.
- ❖ **More Control on Hardware:** Assembly language provides more control over hardware than high-level languages by directly interfacing with the processor's instruction set, memory, and other system resources.

Disadvantages

- ❖ **Machine Dependent:** which means that programs written in these languages are not portable to other, incompatible systems. This makes it a low-level language. If an assembly program is to be shifted to a different type of computer, it has to be modified to suit the new environment.
- ❖ **Harder to Learn:** Being a machine-dependent language, every type of computer architecture requires a different assembly language, making it hard for a programmer to remember and understand every dialect of assembly.
- ❖ **Slow Development Time:** Assembly generated applications are slower to develop as compared to high-level-language- based applications.
- ❖ **Less Efficient:** A program written in assembly language is less efficient than machine language because every assembly instruction has to be converted into machine language. Therefore, the execution of assembly language program takes more time than a machine language program.
- ❖ **No Standardization:** Assembly languages cannot be standardized because each type of computer has a different instruction set, and therefore, a different assembly language.
- ❖ **No Support for Modern Software Engineering Technology:** Assembly languages provide no inherent support for software engineering technology and safety-critical systems. There is also no specific support for distributed systems. The tools available for working with assembly languages are typically very low-level tools.

List and explain the advantages and disadvantages high level languages

Advantages

- ❖ **Readability:** Since HLL are closer to natural languages, they are easier to learn and understand. In addition, a programmer does not need to be aware of computer

architecture; even a lay person can use it without much difficulty. This is the main reason of HLL's popularity.

- ❖ **Machine Independent:** HLL are machine independent in the sense that a program created using HLL can be used on different platforms with very little or no change at all.
- ❖ **Easy Debugging:** In HLL programmers can concentrate on finding the solution to the problem rapidly, rather than on low-level details of data representation, which results in fewer errors. Moreover, the compilers and interpreters are designed in such a way that they detect and point out the errors instantaneously.
- ❖ **Easier to Maintain:** As compared to low-level languages, the programs written in HLL are easy to Modify and maintain because HLL programs are easier to understand.
- ❖ **Easy Documentation:** Since the statements written in HLL are similar to natural languages, they are easier to understand as compared to low-level languages.
- ❖ **Low Development Cost:** HLL permit faster development of programs. Although a high- level program may not be as efficient as an equivalent low- level program, but the savings in programmer's time generally outweighs the inefficiencies of the application

Disadvantages

- ❖ **Poor Control on Hardware:** HLL are developed to ease the pressure on programmers so that they do not have to know the intricacies of hardware. As a result, developers have less control over system resources and hardware, which can be a disadvantage when performance optimization is critical
- ❖ **Less Efficient:** The HLL applications are less efficient as far as computation time is concerned. HLL must be created and sent through another processing program known as a compiler. This process of translation increases the execution time of an application. Programs written in HLL take more time to execute, and require more memory space.

Explain Fourth Generation Languages.

- ❖ Fourth generation languages (4GLs) have simple, English-like syntax rules, commonly used to access databases.
- ❖ They are a type of high-level, non-procedural programming language designed to be easier to use and more productive than lower-level languages. The 4GL's are divided into three categories: **Query Languages, Reports Generator, Application Generator**

Query Languages:

- ❖ They allow the user to retrieve information from databases by following simple syntax rules.
- ❖ For example, the database may be requested to locate details of all employees drawing a salary of more than \$10,000. Structured Query Language (SQL) and IBM's Query-By-Example (QBE) are examples of query languages.

Report Generators:

- ❖ A report generator is a computer program that automates the creation of structured documents by extracting, processing, and presenting data from various sources like databases, spreadsheets, or other systems in human-readable formats such as PDF, CSV (excel), or HTML.

Application Generators:

- ❖ With application generators, the user writes programs to allow data to be entered into the database. The program prompts the user to enter the needed data. It also checks the data for validity. Cincom System's MANTIS and ADS by Cullinet are examples of application generators.

Advantages of 4GLs:

- ❖ The main advantage of 4GLs is that a user can create an application in a much shorter time for development and debugging than with other programming languages.
- ❖ It does not require the programmers to provide the logic to perform a task. As a result, lot of programming effort is saved.

Disadvantages of 4GLs:

- ❖ Since programs written in a 4GL are quite lengthy, they need more disk space and a large memory capacity as compared to 3GLs.
- ❖ These languages are inflexible also because the programmer's control over language and resources is limited as compared to other languages.
- ❖ These languages cannot directly utilize the computer power available at the hardware level as compared to other levels of languages.

Write an algorithm to display maximum of three numbers

Step 1: START

Step 2: INPUT A, B, C

Step 3: If (A>B) and (A>C) then

 print "A is greater".

Else if (B>A) and (B>C) then

 print "B is greater".

Else

 print "C is greater".

Step 4: STOP

Write an algorithm to calculate the factorial of a number

Step 1: START

Step 2: INPUT N

Step 3: Fact = 1, i=1

Step 4: Repeat for i=1 to n
 Fact=Fact*i

Step 5: Print Fact

Step 6: STOP

List and Explain Features of Good Programming Language.

Ease of Use:

- ❖ A good programming language must be simple and easy to learn and use.
- ❖ It should provide a clear, simple, and agreeable set of concepts to a programmer that they can grasp quickly. Simplicity helps in the readability of the language.

Portability:

- ❖ The language should support the construction of code in a way that it could be distributed across multiple platforms (operating systems), programs written on one platform should be able to be tested or transferred to any other computer or platform and there it should perform accurately.

Reliability:

- ❖ Reliability is concerned with making a system failure free, and thus is concerned with all possible errors.
- ❖ The language should provide support for preventing errors. For example, some errors can be prevented by strict syntax checking. Apart from prevention, the language should also be able to detect and report errors in the program.

Safety

Safety tries to ensure that any failure that results in minor consequences, and even potentially dangerous failures are handled in a fail-safe fashion. Language can facilitate this through such features as built-in consistency checking and exceptional handling. Ex: application loses connection to the bank's server in such situation proper instruction should be provided to the user.

Maintainability:

Creating an application is not the end of the system development. It should be maintained regularly so that it can be easily modified to satisfy new requirements or to correct deficiencies.

Cost:

Cost component is a primary concern before deploying a language at a commercial level. It includes several costs such as the following.

- ❖ Program execution and translation cost.
- ❖ Program creation, testing and usage cost.
- ❖ Program maintenance cost.

Compact Code:

- ❖ A language should promote compact coding, that is, the intended operations should be coded in a minimum number of lines. Even if the language is powerful, and is not able to perform the tasks in a small amount of code, then it is of no use.
- ❖ The main reason behind it is that a large code requires more testing and developing time, thereby increasing the cost of developing an application.

Naturalness for the Application:

- ❖ The language should have a syntax which allows the program structure to show the underlying logical structure of algorithm. A programming language should provide a conceptual framework for thinking through algorithms and means of expressing those algorithms through flowcharts.

Reusability:

- ❖ The language should facilitate the adaptation of code for use in other applications. Code is reusable when it is independent of other codes. It is very common, for example, to reuse common data structures such as stacks, queues and trees. When these have been defined with common operations on the structures, these abstract data types are easy to reuse.

Discuss flowcharts. List and explain the benefits of flowcharts

- ❖ A flowchart is a pictorial representation of an algorithm in which the steps are drawn using different geometric shapes. Each geometric shape represents an operation. The logical flow is indicated by interconnecting arrows.
- ❖ The primary purpose of the flowchart is to help the programmer in understanding the logic of the program.

Benefits of Flowcharts

- ❖ **Makes Logic Clear:** The main advantage of using a flowchart to plan a task is that it provides a pictorial representation of the task, which makes the logic easier to follow. The symbols are connected in such a way that they show the movement (flow) of information through the system visibly. The steps and how each step is connected to the next can be clearly seen.
- ❖ **Communication:** Being a graphical representation of a problem-solving logic, flowcharts are a better way of communicating the logic of a system to all concerned.
- ❖ **Effective Analysis:** With the help of a flowchart, the problem can be analyzed in an effective way. This is because the analyzing duties of the programmers can be delegated to other persons, who may or may not know the programming techniques, as they have a broad idea about the logic.

- ❖ **Useful in Coding:** The flowcharts act as a guide or blueprint during the analysis and program development phase. Once the flowcharts are ready, the programmers can plan the coding process effectively as a result, error-free programs are developed in HLL.
- ❖ **Proper Testing and Debugging:** By nature, a flowchart helps in detecting the errors in a program, as the developers know exactly what the logic should do.
- ❖ **Appropriate Documentation:** Flowcharts serve as a good program documentation tool. Since normally programs are developed for novice users, they can take the help of the program documentation to know what the program actually does and how to use the program.

Discuss flowcharts. List and explain the limitations of flowcharts

Some of the limitations of the flowchart are given below.

- ❖ **Complex:** The major disadvantage in using flowcharts is that when a program is very large, the flowcharts may continue for many pages, making them hard to follow. Flowcharts tend to get large very quickly and it is difficult to follow the represented process.
- ❖ **Costly:** If flowcharts are to be drawn for a huge application, the time and cost factor of program development may get out of proportion, making it a costly affair.
- ❖ **Difficult to Modify:** Due to its symbolic nature, any changes or modification to a flowchart usually requires redrawing the entire logic again, and redrawing a complex flowchart is not a simple task.
- ❖ **No Update:** Usually programs are updated regularly. However, the corresponding update of flowcharts may not take place, especially in the case of large programs. As a result, the logic used in the flowchart may not match with the actual program's logic.

Discuss pseudocode. List and explain the advantages and limitations of pseudocode

- ❖ Pseudocode is an outline of a program, written in a form that can be easily converted into actual programming statements.
- ❖ Pseudocode uses plain English statements rather than symbols to represent the processes in a computer program. It is also known as PDL (program design language), as it emphasizes more on the design aspect of a computer program or structured English, because usually pseudocode instructions are written in normal English, but in a structured way.

Benefits of Pseudocode

- ❖ Since it is language independent, it can be used by most programmers. It allows the developer to express the problem logic in

plain natural language.

- ❖ It is easier to develop a program from a pseudocode rather than from a flowchart or decision table.
- ❖ Programmers do not have to think about syntaxes; they simply have to concentrate on the underlying logic. The focus is on the steps to solve a problem rather than on how to use the computer language.
- ❖ The words and phrases used in pseudocode are in line with basic computer operations. This simplifies the translation from the pseudocode to a specific programming language.
- ❖ Unlike flowcharts, pseudocode is compact and does not tend to run over many pages.
- ❖ It's simple structure and readability makes it easier to Modify.

Limitations of Pseudocode

Some of the most notable limitations are as follows.

- ❖ It does not provide a visual representation of the program's logic.
- ❖ There are no accepted standards for writing pseudocodes. Programmers use their own style of writing pseudocode.
- ❖ It is quite difficult for the beginners to write pseudocode as compared to drawing a flowchart.

Explain the purpose of Pseudocode. List and explain keywords commonly used in pseudocode

The goal of writing pseudocode is to provide a high-level description of an algorithm, which facilitates analysis and eventual coding, but at the same time suppresses many of the details that are insignificant.

Pseudocode uses some **keywords** to denote programming processes. Some of them are as follows.

- **Input:** READ, OBTAIN, GET and PROMPT
- **Output:** PRINT, DISPLAY and SHOW
- **Compute:** COMPUTE, CALCULATE and DETERMINE
- **Initialize:** SET and INITIALIZE
- **Add One:** INCREMENT

Pseudocode: To Calculate the Area of A Rectangle

- ❖ PROMPT the user to enter the height of the rectangle
- ❖ PROMPT the user to enter the width of the rectangle COMPUTE the area by multiplying the height with width
- ❖ DISPLAY the area STOP